Full Stack Assignment – 36

Assignment ID: FSD-36

Title: Time-Based Access Window System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Time-Based Access Window System** that allows administrators to grant users access only within a defined time window.

The system must enforce time-based access validation and ensure **data persistence using a database**.

2️⃣ User Roles

USER

- Register and log in to the system
- View access status
- Attempt to access protected resources

ADMIN

- Register and log in to the system
- Define access start and end times for users
- View access logs

3️⃣ Functional Requirements

Authentication & Authorization

- User registration and login
- **JWT-based authentication**
- Protected backend APIs
- Role-based access control

Access Window Management

- ADMIN sets access start & end timestamps for a user
- Access window data must be stored in the database
- Access checks must validate current server time

4️⃣ Business Rule (Critical)

⚠️ **Users can access the system only within their assigned time window.**

Access attempts outside the window must be rejected.

5 Technical Constraints

- All access data **must persist in a database**
- No hardcoded or in-memory logic
- JWT and role validation mandatory
- Environment variables required

6 Frontend Requirements

- Login & registration pages
- Role-based dashboards
- Access status display for USER
- Access window configuration for ADMIN

7 Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **revocation enforcement, role control, and end-to-end execution**.